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Social Network Analysis of a Public Facebook Sample

A simple guide to the data, method and results

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Academic project report

Abstract

We used Scrapling to collect a small public Facebook sample and turn observed comments and tags into a network. The project includes the data, Python scripts, a notebook, figures and graph files. The main 2026 network has 27 profiles and 9 directed connections. It describes the records we could see, not all activity in the groups.

Collect evidence	Validate records	Construct graphs	Report results

1. What we collected

The data was collected on 13 October 2026 from publicly visible Facebook pages. The complete sample contains the following records:

Groups	Posts	Comments	Profiles	Tags	Profile-group pairs
7	63	25	73	5	71

Each post has an exact post ID, author ID, group ID and publication time. IDs connect records across the tables. The sample contains 66 served User profiles and 7 group-scoped anonymous profiles. A profile does not establish a verified real-world identity.

The main question is: **what commenting and tagging connections did we observe in WEBC and Expats in Warsaw during 2026?** The time window is 1 January through 13 October 2026, using Europe/Warsaw dates.

2. How we used Scrapling



Scrapling
Python scraping framework used for the public collection.

Step 1 - Open public content. Scrapling loaded group previews and direct post links without logging in. Its browser fetcher captured the pages and the structured responses those pages requested.

Step 2 - Extract records. The Python tools read IDs, authors, groups, publication times, comments and explicit tags from the returned data. They saved the results as CSV tables.

Step 3 - Check the data. We removed duplicate records and checked IDs, links between tables and timestamps. Missing information was left blank. We did not invent users, comments or connections.

Public access was incomplete. A login wall or a limited preview could hide records. A successful page request did not mean that all comments were available.

3. How the network works

Node = a profile. A### labels make the drawing readable. Their mapping is stored in node_mapping.csv. Anonymous profiles stay tied to their group; we do not infer who they are.

Comment arrow: comment author to post author. This is the chosen analysis rule; it does not prove that the comment directly addresses that person.

Tag arrow: content author to the explicitly tagged profile. A written name without an ID does not create an edge.

Weight = number of observed events. Repeated events between the same two profiles increase the weight. Self-comments stay in the source data but are removed from the analysis graph.

Isolate = a profile without a recovered connection. Isolates remain in the totals and graph files. The drawing shows only profiles with connections.

A separate profile-group graph records observed posting and commenting. It is not a membership or friendship list.

4. The main result

Sample	Profiles	Connections	Isolates
All 7 groups / all years	73	26	43
Two community groups / all years	31	12	17
Two community groups / 2026	27	9	17

The main 2026 sample contains **24 posts, 6 comments and 3 tags**. Its network has **27 profiles, 9 directed connections and 17 isolates**. Only 10 profiles have a recovered connection. The largest connected group contains 7 profiles when arrow direction is ignored.

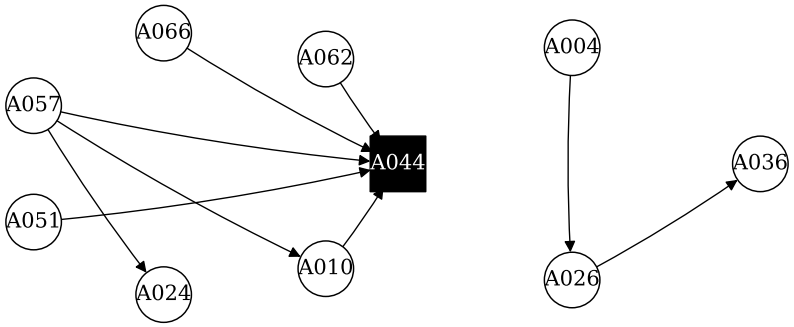


Figure 1. Main 2026 network. Open circles: served User profiles. Solid squares: group-scoped anonymous profiles. The 17 isolates remain in the graph files and totals.

The project also calculates degree (number of connections), strength (number of events), density (fraction of possible connections), components (connected sets) and betweenness (shortest-path position). These are sample descriptions, not measures of real-world influence.

For the 2026 variant, both the parent post and the comment/tag time must fall inside the window. The larger variants add other sources or older content, so their totals do not show changes over time.

5. What these results can tell us

Enough for a minimum exploratory SNA. We can build a network from exact records, describe its size and connections, and compare three clearly defined samples.

Too incomplete for whole-community claims. Comments are only partly visible, no verified nested replies were collected, and the sources were not chosen randomly. Missing edges do not prove missing real relationships.

We cannot infer complete membership, friendships, influence, or the identities of all reacting and sharing users. Engagement totals are repeated snapshots; adding them together would double-count activity.

A stronger next study should use the same groups and time window, document attempted and recovered records, and collect more comparable comments and verified replies.

6. How to run the project

Step 1. Open the project folder and install the packages using Python 3.13, the tested version:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

Step 2. Rebuild the analysis, check the outputs and regenerate this report:

```
python analysis.py
python validate_project.py
python generate_report.py
```

Step 3. Open analysis.ipynb, choose the same Python environment and run all cells for the explained walkthrough. The delivered notebook already contains successful outputs. All 29 project validation checks passed.

The frozen inputs are in data/. The metrics and graph files are in results/, and charts are in figures/. README.md explains all files; DATA_DICTIONARY.md explains the fields.

This workflow runs offline. Optional new scraping is separate: collect_data.py saves fresh captures in collections/ without changing the frozen data. See README.md for its commands and collection limits.

References and links

Project evidence: frozen CSV tables in data/, DATA_DICTIONARY.md, results/summary.json and results/validation.json. The URLs below are provided together for source attribution and further reading. Retrieved 13 October 2026.

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[18] Impreza i Zabawa, to Warszawa! - observed Facebook source
https://www.facebook.com/groups/tandem.evening/

[19] Wspieraj lokalny biznes: KRAKÓW - observed Facebook source
https://www.facebook.com/groups/639972129911751/

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